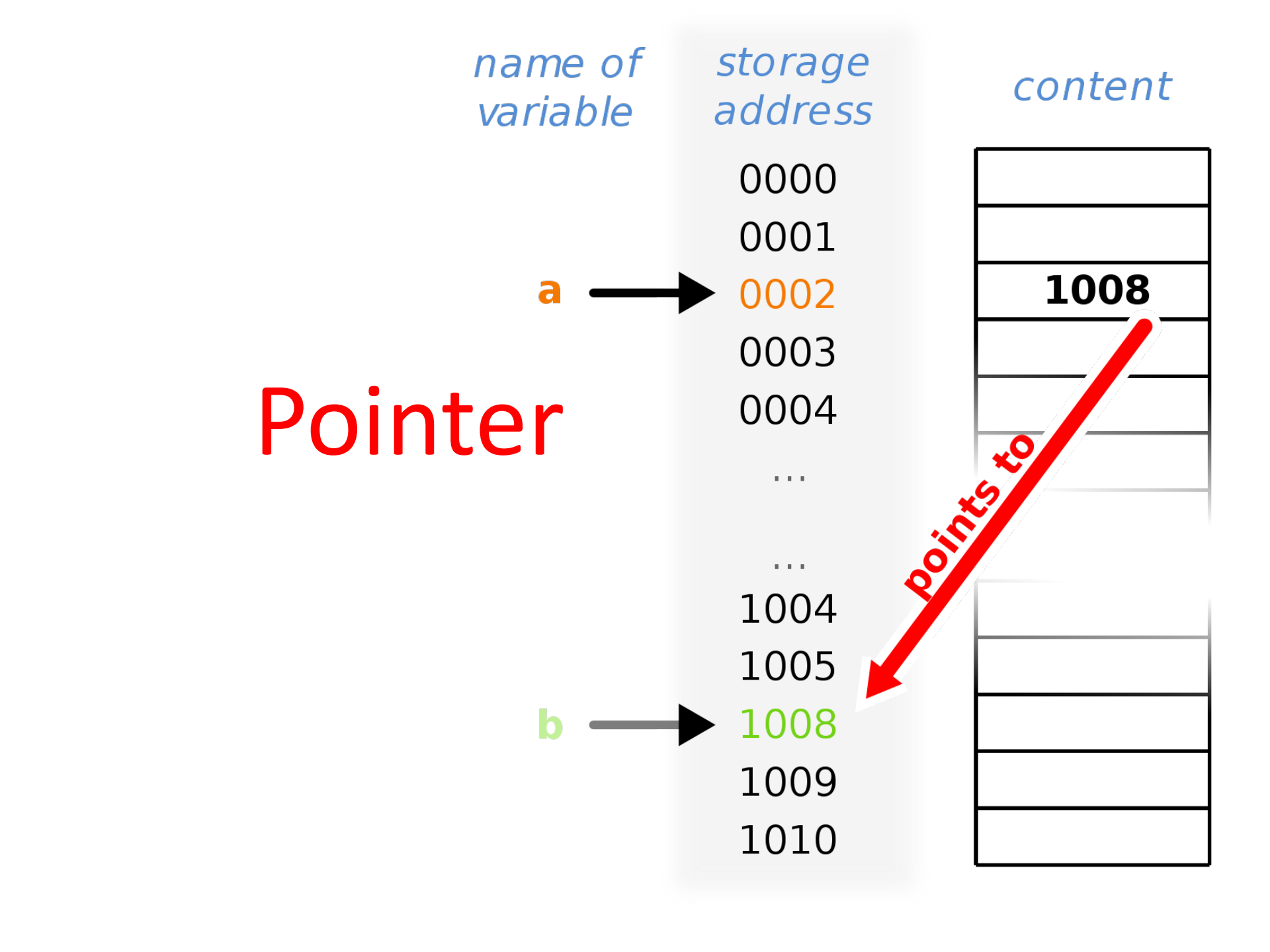
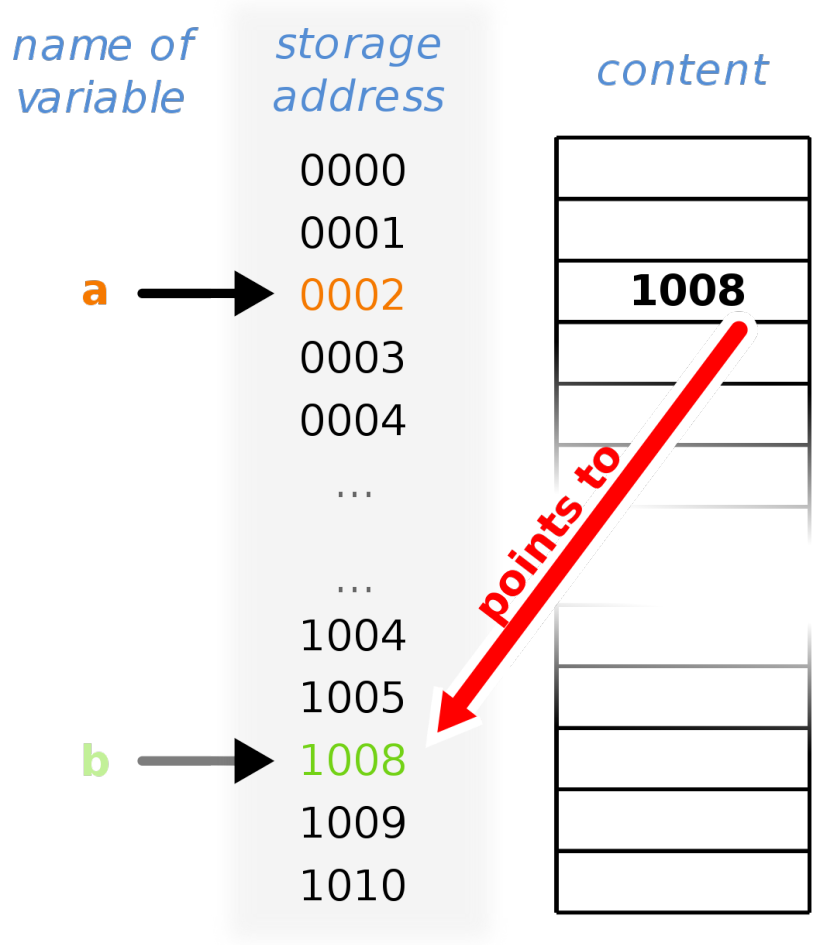

name of variable
storage address
content
Pointer
a
b
0000
0001
0002
0003
0004
...
...
1004
1005
1008
1009
1010
1008
points to

What is Pointer?

❖ A **pointer** is a variable whose value is the address of another variable.

- Computer memory is divided into **byte** for addressing.
- Each byte is numbered sequentially- which is called **storage address**.
- Storage address is represented in **hexadecimal** number system. In hexadecimal, following symbols are used: **A for 10**, B for 11, **C for 12**, D for 13, **E for 14** and F for 15.



What is Pointer?

Declaration:

int *a, b;

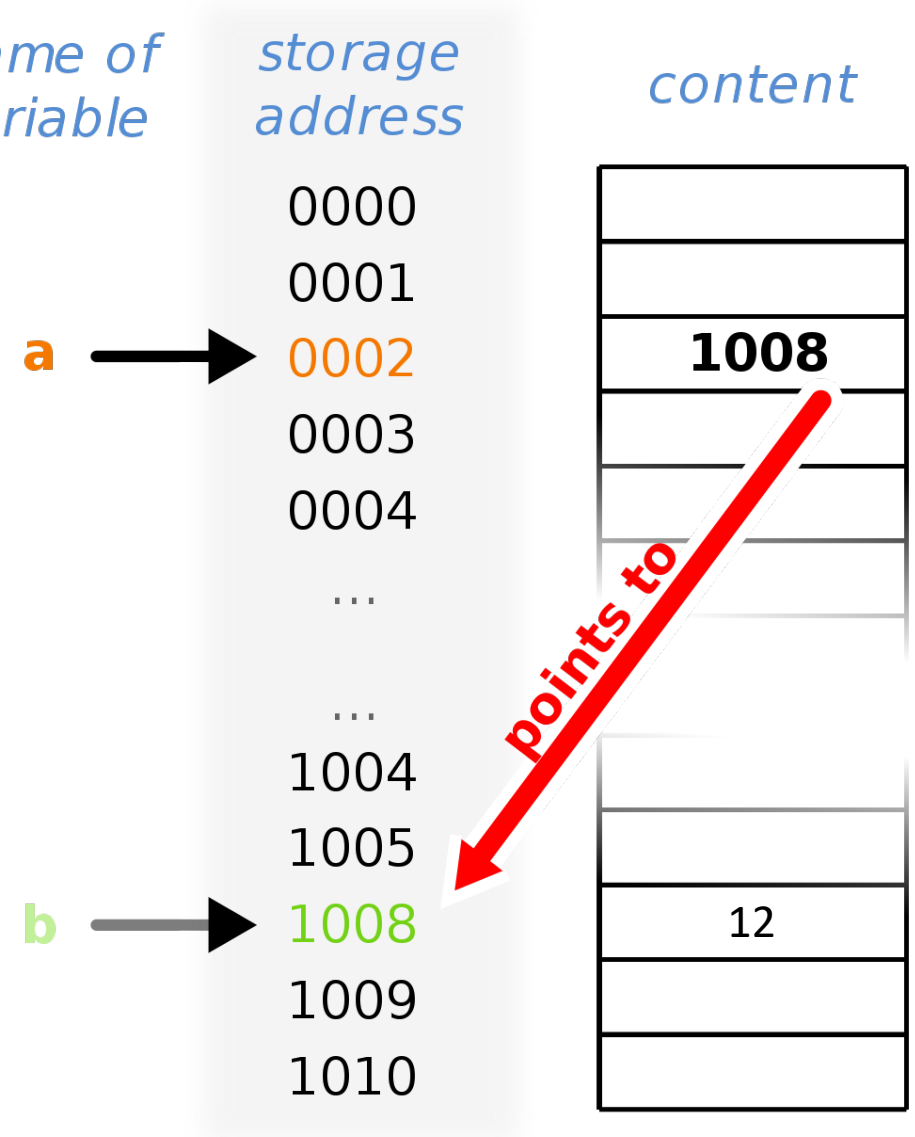
Understanding point:

Question	Answer
&a	0002
a	1008
*a	12
&b	1008
b	12

Read out:

*a: pointer of a

&a: ampersand or address of a

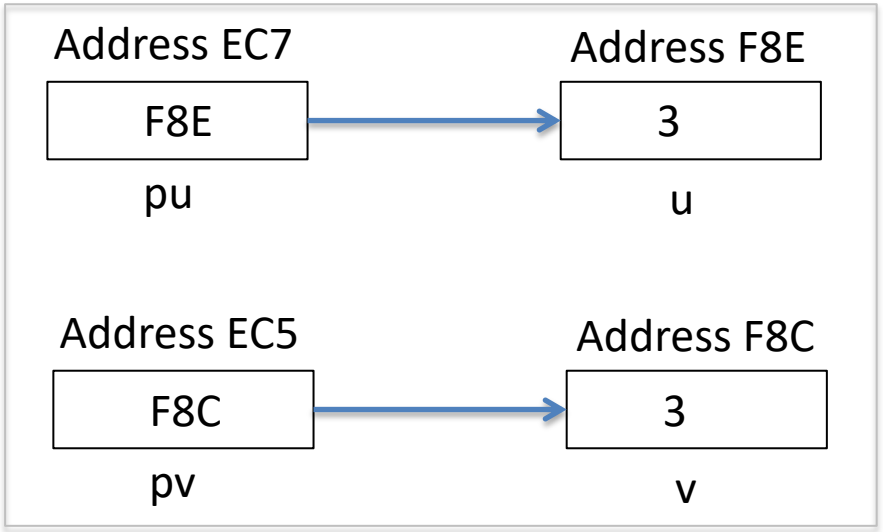


Pointer Examples 1

Write down the output of the following program:

```
#include <stdio.h>
main(){
      int u = 3, v, *pu, *pv;

      pu = &u;
      v = *pu;
      pv = &v;

      printf("\nu=%d    &u=%x   pu=%x    *pu=%d", u, &u, pu, *pu);
      printf("\nv=%d    &v=%x   pv=%x    *pv=%d", v, &v, pv, *pv);
}
```

Assume that, &pu = EC7, &u = F8E, &pv = EC5, &v = F8C



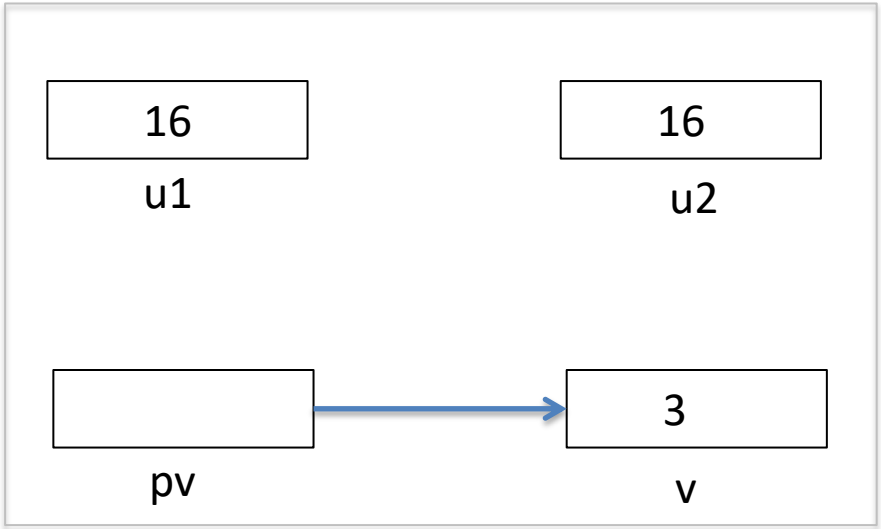
Output

u = 3 &u =F8E pu =F8E *pu = 3
v = 3 &v =F8C pv =F8C *pv = 3

Pointer Examples 2

Write down the output of the following program:

```
#include <stdio.h>
main(){
        int u1, u2;
        int v = 3, *pv;

        u1 = 2 * ( v + 5 );
        pv = &v;
        u2 = 2 * ( *pv + 5 );

        printf("\nu1=%d     u2=%d", u1, u2);
}
```



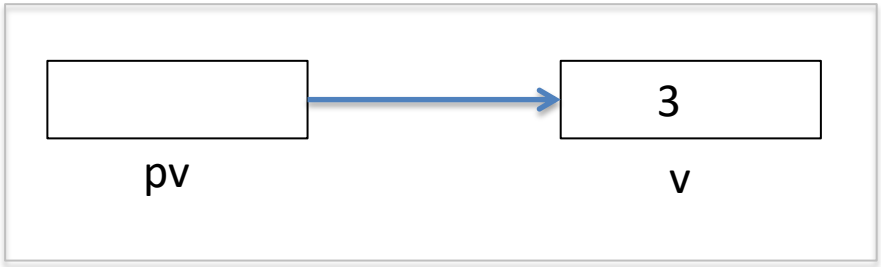
Output

u1 = 16 u2 =16

Pointer Examples 3

Write down the output of the following program:

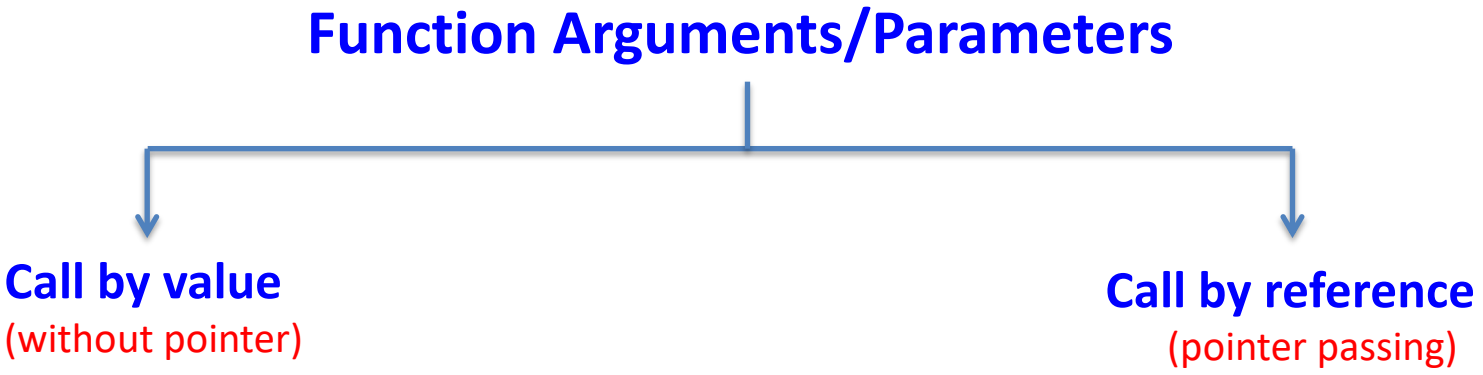
```
#include <stdio.h>
main(){
        int v = 3;
        int *pv;

        pv = &v;
        printf("\n*pv=%d     v=%d", *pv,  v);


        *pv = 0;
        printf("\n*pv=%d     v=%d", *pv,  v);
}
```



Output

*pv = 3 v = 3
*pv = 0 v = 0

Passing Pointer to Function



- **Call by value**: passing the value of a variable to a function as an argument.

```
int funct(int u, int v);
```

- **Call by reference**: passing the address of a variable to a function as an argument.

```
int funct(int *pu, int *pv);
```

Call by value & call by reference Example

Write down the output of the following program:

```
#include <stdio.h>

void funct1(int u, int v);
void funct2(int *pu, int *pv);

main(){
      int u = 1, v = 3;

      printf("\nBefore calling funct1:   u = %d    v = %d", u,  v);
      funct1(u, v);
      printf("\n\n\nAfter calling funct1:   u = %d    v = %d", u,  v);

      printf("\nBefore calling funct2:   u = %d    v = %d", u,  v);
      funct2(&u, &v);
      printf("\nAfter calling funct2:   u = %d    v = %d", u,  v);
}
```

Call by value & call by reference Example Continue..

```
void funct1(int u, int v){
        u = 0;
        v = 0;
        printf("\nWithin funct1:     u = %d      v = %d", u, v);
}
```


```
void funct2(int *pu, int *pv){
        *pu = 0;
        *pv = 0;
        printf("\nWithin funct2:     *pu = %d      *pv = %d", *pu, *pv);
}
```

Call by value & call by reference Example Continue..

Output:

Before calling funct1:	u = 1	v = 3
Within funct1:	u = 0	v = 0
After calling funct1:	u = 1	v = 3
Before calling funct2:	u = 1	v = 3
Within funct2:	*pu = 0	*pv = 0
After calling funct2:	u = 0	v = 0

Pointer and Array

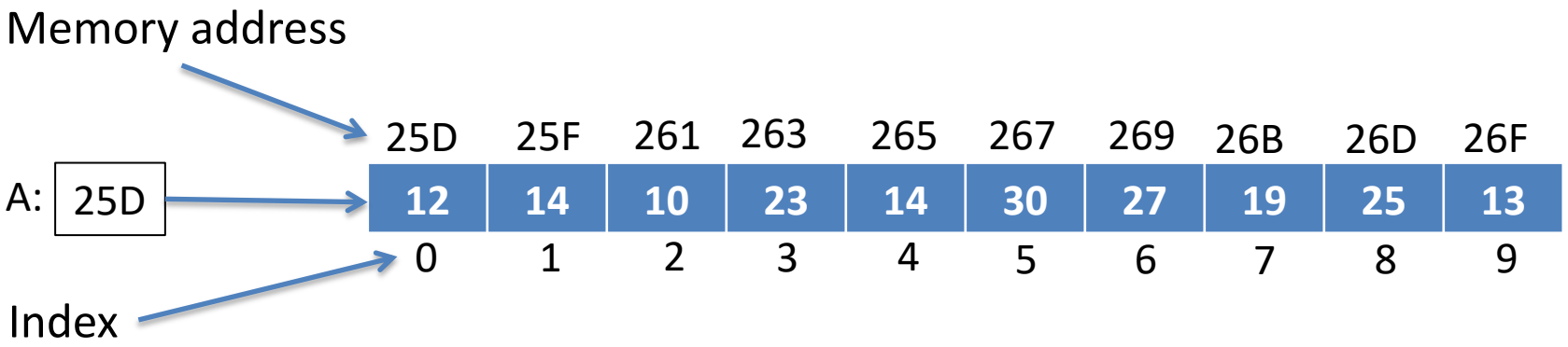
- An **array name** is a variable which points to some consecutive memory location.

Declaration of 1D Array:

int A[10];

Required memory size:

char	1 byte
int	2 bytes
float	4 bytes



Example:

A[0] = 12	A[3] = 23	A[7] = 19
*A = 12	*(A + 3) = 23	*(A + 7) = 19
A = 25D	(A + 3) = 263	(A + 7) = 26B

Pointer and Array

Example:

For **char** pointer, char *A;

A

A +1 = A + size of **char**

A + 5 = A + 5 * size of **char**

Memory size:

char	1 byte
int	2 bytes
float	4 bytes

For **float** pointer, float *A;

A

A +1 = A + size of **float**

A + 2 = A + 2 * size of **float**

For **int** pointer, int *A;

A

A +1 = A + size of **int**

A + 3 = A + 3 * size of **int**

Pointer and Array

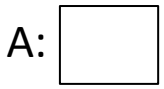
Declaration of 1D Array:

int A[10];

Declaration of 1D Array using Pointer:

step 1. Declaration of pointer

int *A;

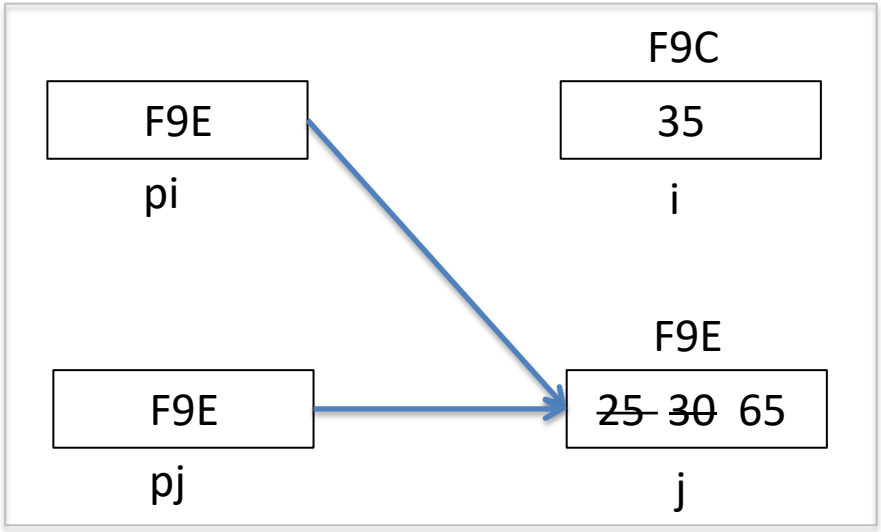


step 2: Allocation of Memory

A = (int *) malloc(10*sizeof(int));

Pointer Exercise 1

```
int i, j = 25;
int *pi, *pj = &j;
……..
*pj = j + 5;
i = *pj + 5;
Pi = pj;
*pi = i + j;
```



The value assigned to i begins at address F9C and the value assigned to j begins at address F9E.

Output

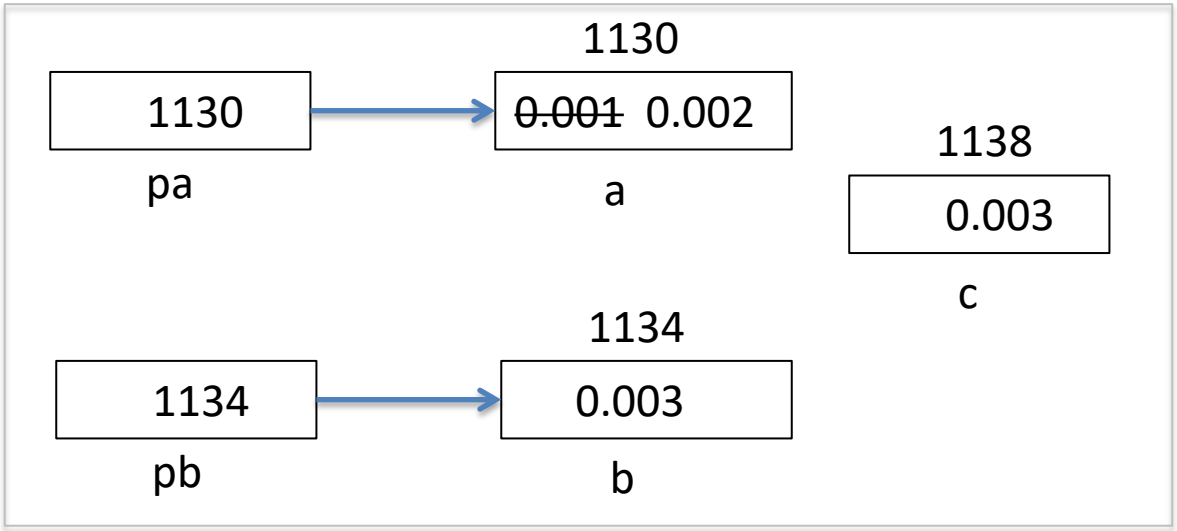
&i = F9C &j = F9E pj = F9E *pj = 65

i = 35 pi = F9E *pi = 65 (pi + 2) = F9E + 4 = FA2

*(pi + 2) = unknown.

Pointer Exercise 2

```
float a = 0.001, b = 0.003;
float c, *pa, *pb;
pa = &a;
*pa = 2 * a;
pb = &b;
c = 3 * (*pb - *pa);
```



The value assigned to a begins at address 1130 and the value assigned to b begins at address 1134 and the value assigned to c begins at 1138.

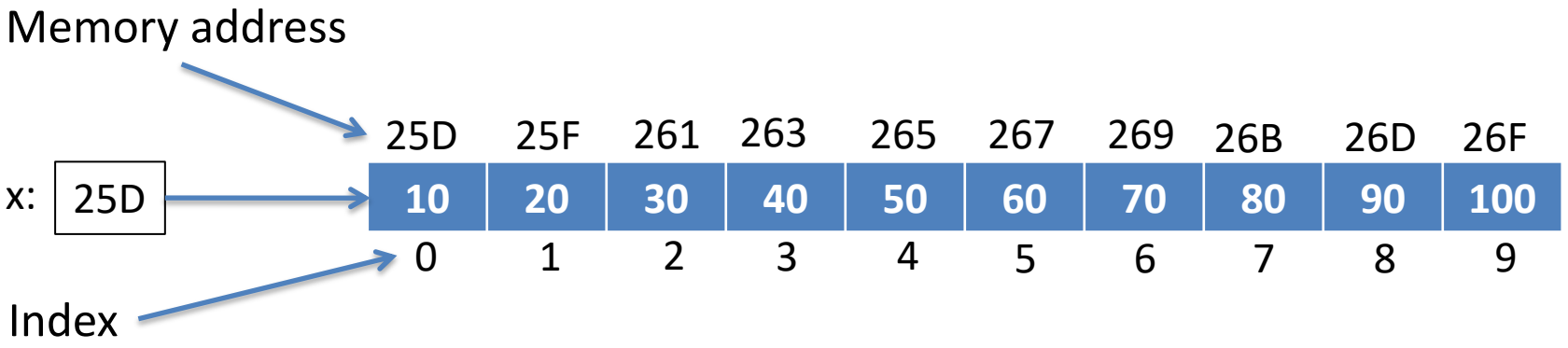
Output

&a = 1130	&b = 1134	&c = 1138
pa = 1130	*pa = 0.002	&(*pa) = 1130
pb = 1134	*pb = 0.003	c = 0.003

Pointer Exercise 3

static int x[10] = { 10, 20, 30, 40, 50, 60, 70, 80, 90, 100 }

The value assigned to x begins at address 25D.



Output

x = 25D

(x + 2) = 25D + 2*2 = 261

*x = 10

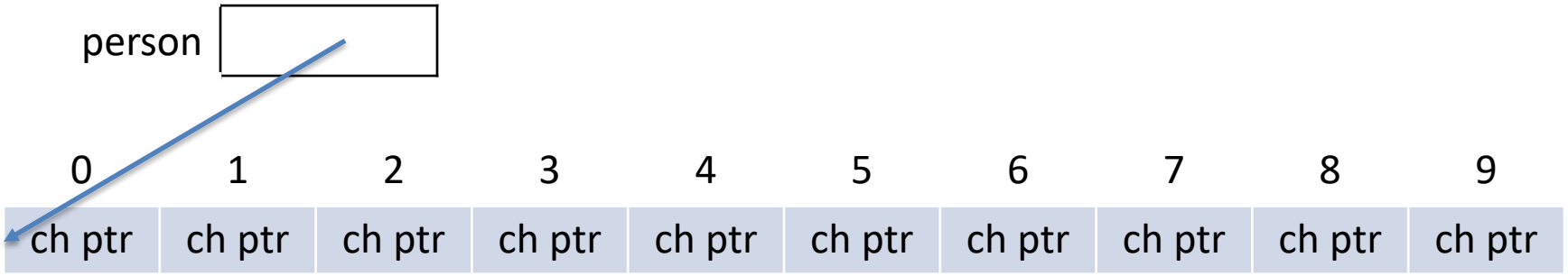
(*x + 2) = 10 + 2 = 12

*(x + 2) = 30

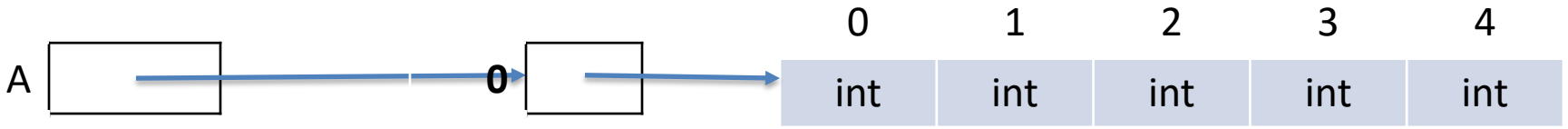
Pointer and 2D Array

- *** operator** has a lower precedence than the **[] operator.**

Declaration 1: **char *person[10];** //Array of pointers



Declaration 2: **int (*A)[5];** //Pointer to an array

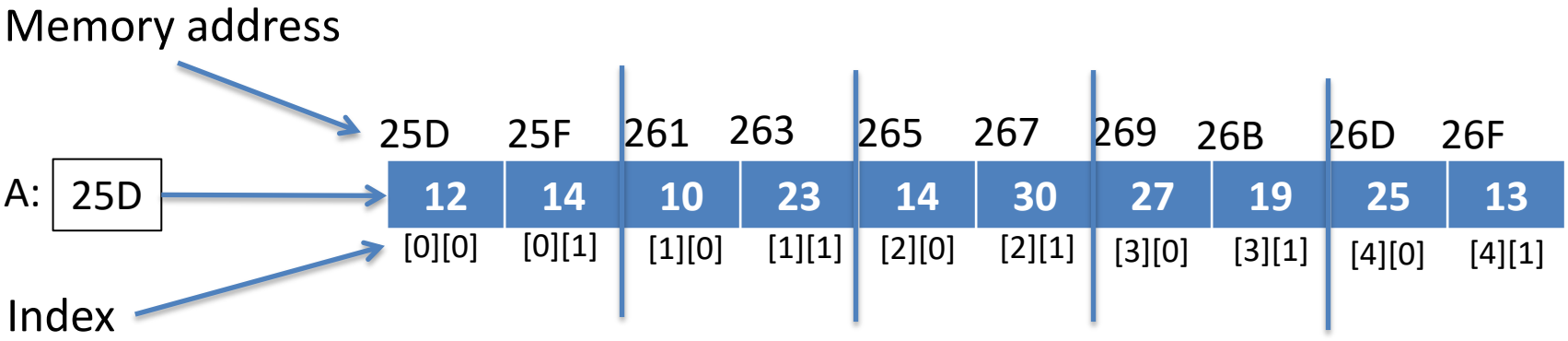


- This is useful for processing two-dimensional array parameters declared with **unknown number of rows**.

Pointer and 2D Array

Declaration of 1D Array:

int A[5][2];



Example: A[2][1] = 30 *(*(A+2) +1) = 30

A = 25D, *A = 25D, **A = 12

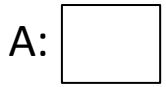
(A+1) = 261, *A + 1 = 25F, *(A+1) = 261

Pointer and 2D Array

Declaration of 2D Array using Pointer:

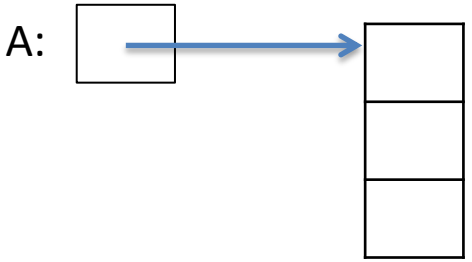
step 1. Declaration of pointer

```
int  **A;
```



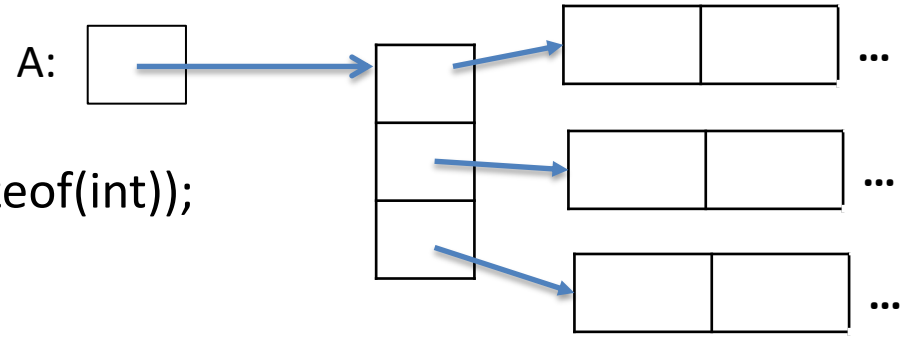
step 2: Allocation of row

```
A = (int *) malloc(row*sizeof(int *));
```



step 3: Allocation of Memory

```
for ( i = 0; i < row; ++i)
  A[i] = (int) malloc(col*sizeof(int));
```



Pointer and function

int *func_name (int a); //function returns a pointer to an integer

int (*func_name)(int a); // pointer to a function

Try Yourself…..

(1) int *(*f)(int a);

(2) int *(*p)(int (*a)[]);

(3) int *(*p)(int *a[]);

(4) char *f(char *a);